T1 1D $\because g: R \to R$ is a convex and nondecreasing function. $h: R^n \to R$ is a convex function.

Assume $\lambda \in [0,1]$

$\therefore f(\lambda x + (1-\lambda) y) = g(h(\lambda x + (1-\lambda) y))$

$\because h(\lambda x + (1-\lambda) y) \le \lambda h(x) + (1-\lambda) h(y)$

and g is a nondecreasing function

$\therefore g(h(\lambda x + (1-\lambda) y)) \le g(\lambda h(x) + (1-\lambda) h(y))$

$\le \lambda g(h(x)) + (1-\lambda) g(h(y))$

$\therefore f(x) = g(h(x))$ is a convex function.

(2) $\because h$ is a concave function

$\therefore -h$ is a convex function.

$\therefore -h(\lambda x + (1-\lambda) y) \le -\lambda h(x) - (1-\lambda) h(y)$

$\Rightarrow h(\lambda x + (1-\lambda) y) \ge \lambda h(x) + (1-\lambda) h(y)$.

$\therefore f(\lambda x + (1-\lambda) y) = g(h(\lambda x + (1-\lambda) y)) \le g(\lambda h(x) + (1-\lambda) h(y)) \le \lambda g(h(x)) + (1-\lambda) g(h(y))$

$\therefore f(x) = g(h(x))$ is a convex function.

(3) when $h(x) = Ax + b$.

$\therefore h(\lambda x + (1-\lambda) y) = A(\lambda x + (1-\lambda) y) + b = \lambda(Ax + b) + (1-\lambda)(Ay + b)$

$= \lambda h(x) + (1-\lambda) h(y)$

$\therefore f(\lambda x + (1-\lambda) y) = g(h(\lambda x + (1-\lambda) y)) = g(\lambda h(x) + (1-\lambda) h(y))$

$\le \lambda g(h(x)) + (1-\lambda) g(h(y))$

$\therefore f(x) = g(h(x))$ is a convex function.

T2 $f(x_1, x_2) = e^{2x_1^2 - x_2^2} - 3x_1 + 5x_2$

$\therefore f_1' = 4x_1 e^{2x_1^2 - x_2^2} - 3$; $f_2' = -2x_2 e^{2x_1^2 - x_2^2} + 5$

$f_{11}'' = 4e^{2x_1^2 - x_2^2} + 4x_1 \cdot (4x_1) \cdot e^{2x_1^2 - x_2^2}$

$= (4 + 16x_1^2) e^{2x_1^2 - x_2^2}$

$f_{22}'' = -2e^{2x_1^2 - x_2^2} - 2x_2 \cdot (-2x_2) e^{2x_1^2 - x_2^2}$

$= (4x_2^2 - 2) e^{2x_1^2 - x_2^2}$

$f_{12}'' = 4x_1 \cdot (-2x_2) e^{2x_1^2 - x_2^2} = -8x_1 x_2 e^{2x_1^2 - x_2^2}$

$f(1,1) = e + 2$; $f_1'(1,1) = 4e - 3$; $f_2'(1,1) = -2e + 5$;

$f_{11}''(1,1) = 20e$; $f_{22}''(1,1) = 2e$; $f_{12}''(1,1) = -8e$

$\therefore f(x_1, x_2) = f(1,1) + f_1'(1,1)(x_1 - 1) + f_2'(1,1)(x_2 - 1)$

$= (e+2) + (4e-3)(x_1 - 1) + (-2e+5)(x_2 - 1)$

is the linear approximation.

$f(x_1,x_2) = (e+2)+(4e-3)(x_1-1)+(-2e+5)(x_2-1)+\frac{1}{2}f_{11}''(x_1-1)^2+\frac{1}{2}f_{22}''(x_2-1)^2+2\cdot\frac{1}{2}f_{12}''(x_1-1)(x_2-1)$

$=(e+2)+(4e-3)(x_1-1)+(-2e+5)(x_2-1)+10e(x_1-1)^2+e(x_2-1)^2-8e(x_1-1)(x_2-1)$.

Because of linear approxiamation is the linear function, neither.

for quadratic approxiamation, the Hessian matrix:

$$H=\begin{pmatrix} f_{11}'' & f_{12}'' \\ f_{21}'' & f_{22}'' \end{pmatrix}=\begin{pmatrix} 20e & -8e \\ -8e & 2e \end{pmatrix} \quad \because 20e, 2e>0$$

$20e\cdot 2e-(-8e)^2=40e^2-64e^2=-24e^2<0$

$\therefore$ for quadratic approxiamation, neither.

T3 Assume that $f(x_1,x_2)=2x_1^2+x_2^2-2x_1x_2+2x_1^3+x_1^4$

$x_1\in\mathbb{R}$, $x_2\in\mathbb{R}$.

$\therefore \frac{\partial f}{\partial x_1}=4x_1-2x_2+6x_1^2+4x_1^3$

$\frac{\partial f}{\partial x_2}=2x_2-2x_1$

Let $\frac{\partial f}{\partial x_1}=0$, $\frac{\partial f}{\partial x_2}=0 \Rightarrow 2x_1+6x_1^2+4x_1^3=0=x_1(2+6x_1+4x_1^2)$

$\Rightarrow x_1=0$ or -1 or $-\frac{1}{2}$.

$\therefore$ We have three set of $(x_1,x_2)=(0,0),(-1,-1),(-\frac{1}{2},-\frac{1}{2})$

$\because f_{11}''=4+12x_1+12x_1^2$; $f_{22}''=2$; $f_{12}''=-2$

for $(0,0)$: $H=\begin{pmatrix} 4 & -2 \\ -2 & 2 \end{pmatrix}$, $4,2>0$, $4\times2-(-2)^2=4>0$

$\therefore$ H is positive semidefinite

for $(-\frac{1}{2},-\frac{1}{2})$: $H=\begin{pmatrix} 1 & -2 \\ -2 & 2 \end{pmatrix}$, $1,2>0$, $1\times2-(-2)^2=-2<0$

$\therefore (-\frac{1}{2},-\frac{1}{2})$ neither.

for $(-1,-1)$: $H=\begin{pmatrix} 4 & -2 \\ -2 & 2 \end{pmatrix}$ is positive semidefinite.

$\therefore f(0,0)=0$; $f(-1,-1)=2+1-2+2+1=4$

$\therefore (0,0),(-1,-1)$ is local minimizer.